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Fed-Batch and Perfusion Culture Processes: Economic, Environmental, and Operational Feasibility Under Uncertainty

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ABSTRACT: This article evaluates the current and future potential of batch and continuous cell culture technologies via a case study based on the commercial manufacture of monoclonal antibodies. The case study compares fed-batch culture to two perfusion technologies: spin-filter perfusion and an emerging perfusion technology utilizing alternating tangential flow (ATF) perfusion. The operational, economic, and environmental feasibility of whole bioprocesses based on these systems was evaluated using a prototype dynamic decision-support tool built at UCL encompassing process economics, discrete-event simulation and uncertainty analysis, and combined with a multi-attribute decision-making technique so as to enable a holistic assessment. The strategies were compared across a range of scales and titres so as to visualize how their ranking changes in different industry scenarios. The deterministic analysis indicated that the ATF perfusion strategy has the potential to offer cost of goods savings of 20% when compared to conventional fed-batch manufacturing processes when a fivefold increase in maximum viable cell densities was assumed. Savings were also seen when the ATF cell density dropped to a threefold increase over the fed-batch strategy for most combinations of titres and production scales. In contrast, the fed-batch strategy performed better in terms of environmental sustainability with a lower water and consumable usage profile. The impact of uncertainty and failure rates on the feasibility of the strategies was explored using Monte Carlo simulation. The risk analysis results demonstrated the enhanced robustness of the fed-batch process but also highlighted that the ATF process was still the most cost-effective option even under uncertainty. The multi-attribute decision-making analysis provided insight into the limited use of spin-filter perfusion strategies in industry. The resulting sensitivity spider plots enabled identification of the critical ratio of weightings of economic and operational benefits that affect the choice between ATF perfusion and fed-batch strategies. *Biotechnol. Bioeng.* 2013;110: 206–219.

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KEYWORDS: biopharmaceutical manufacture; antibody production; cell culture; fed-batch; perfusion; process economics; stochastic discrete-event simulation; MySQL database

Introduction

Perfusion culture manufacturing strategies for cell-culture-derived biopharmaceuticals offer the potential of greater daily productivities and hence smaller facility footprints than batch and fed-batch culture manufacturing strategies. However, their use has been hampered historically by perceived greater logistical and validation complexity as well as higher likelihoods of technical failures. More recent perfusion culture systems aim to overcome some of these obstacles with the promise of higher productivities and lower failure rates. This combined with the introduction of single-use technologies for cell culture operations has triggered renewed interest in the potential of bioprocesses based on perfusion culture systems. It is also important to consider less tangible operational factors such as ease of development and flexibility as well as the environmental burden of these strategies. This article focuses on the application of a dynamic decision-support software tool to evaluate how the potential of alternative perfusion culture systems varies across a range of scales and productivities from economic, environmental, and operational perspectives. This is illustrated with a case study that focuses on the impact of different culture strategies on whole bioprocesses, from fermentation through purification, used in the commercial production of monoclonal antibodies (mAbs).

Table I highlights 10 commercial therapeutic biologics that utilize perfusion culture systems for their manufacture. These include recombinant blood factors, enzymes, and mAbs. The choice of perfusion culture has sometimes been

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Table I. Current perfusion cell culture manufacturing strategies.

Product	Protein or mAb	Indication	Company	First approved	Perfusion mode	Reactor size (L)	Reactor number
ReoPro [®]	mAb-abciximab	PCTA	Janssen Biotech**	1994	Spin-filter (internal)	500	5–20 (Leiden)
Cerezyme [®]	Beta-glucocerebrosidase	Gaucher disease	Genzyme	1994	Gravity settler*	2,000	4 (Allston)
Gonal-f [®]	r.FSH	Anovulation	Merck-Serono	1997	Spin-filter*	ND	ND
Remicade [®]	mAb—infliximab	RA + other AI diseases	Janssen Biotech**	1998	Spin-filter (internal)	500	5–20 (Leiden)
					Spin-filter (external)	1,000	8 (Malvern)
Simulect [®]	mAb-basiliximab	Transplant rejection	Novartis	1998	Rotational Sieve filtration	250	ND
Rebif [®]	Interferon beta-1a	Multiple sclerosis	Merck-Serono	1998	Fixed bed	75	22 (Corsier-sur-Vevey)
Kogenate-FS [®]	r.Factor VIII	Haemophilia A	Bayer	2000	Gravity settler	200	8 (Berkeley)
Xigris [®]	r.activated protein C	Sepsis	Eli Lilly	2001	Gravity settler	1,500	2 (CMO)
Fabrazyme [®]	Agalsidase beta	Fabry disease	Genzyme	2003	Gravity settler*	2,000	2 (Allston)
							2 (Framingham)
Myozyme [®]	Alglucosidase alfa	Pompe disease	Genzyme	2006	Gravity settler*	4,000	3 (Geel)
Simponi [®]	mAb—golimumab	RA + other AI diseases	Janssen Biotech**	2009	ATF	1,000	3 (Cork)
						500	5–20 (Leiden)
Stelara [®]	mAb—ustekinumab	Psoriasis	Janssen Biotech**	2009	ATF	500	5–20 (Leiden)

Sources: This data was derived through interviews with industrialists in each company.

ND, not disclosed.

*Microcarrier based process.

**Formerly Centocor.

necessary in cases with labile products (e.g., Xigris[®] [Eli Lilly], Kogenate[®] [Bayer], Cerezyme[®] [Genzyme]). For mAbs historically the choice has been due to company experience and low titres. For example, perfusion culture was the basis of Centocor's (now Janssen Biotech) platform process in the 1980–1990s for both low dose products such as Reopro[®] (30 mg) and high dose products such as the blockbuster Remicade[®] ($\leq 1,050$ mg). However the increase in fed-batch titres combined with their ease of operation has established them as the platform choice for most mAbs in recent years. Table I also reveals that the most common perfusion systems adopted in commercial processes are spin-filters (Deo et al., 1996; Yabannavar et al., 1992) and gravity settlers (often bespoke; Voisard et al., 2003), with up to 4,000 L bioreactors. However, examining the post-launch process changes for Remicade[®] revealed modifications to the perfusion systems used from 500 L internal spin-filter perfusion culture at the Leiden (Netherlands) site to 1,000 L bioreactors with external spin-filters at the Malvern (Pennsylvania) site in an effort to minimize culture failure due to adverse filter fouling (Wojciechowski et al., 2007). In its Ringaskiddy (Cork) site, Janssen Biotech has switched to a newer perfusion technology, alternating tangential flow (ATF) perfusion (Refine Technology, Edison, NJ) for the production of more recent mAbs, such as Simponi[®] (Centocor, 2006). The ATF system achieves media exchange by circulating the broth back and forth between the bioreactor and an external hollow-fiber filter via the action of a diaphragm pump (Crowley et al., 2008; Shevitz, 2000). The vendors report that the pulsating motion allows greater media perfusion rates, reduces the dead volume outside the reactor and minimizes the shear stress the cells are exposed to. These multiple effects allow the ATF perfusion system to

minimize filter fouling, achieve higher cell densities and therefore reach higher productivities compared to earlier perfusion systems.

There are few contributions in the literature that compare the process economics of perfusion and fed-batch manufacturing strategies. Cacciuttolo (2007) concluded that fed-batch culture would be cheaper than perfusion culture as the cost of larger media and harvest tanks in perfusion were considered to outweigh the benefits of higher productivity. The analysis focused on the changes to the cell culture stages rather than the whole bioprocess. Bosch et al. (2008) took into account the complete bioprocess from seed train to final bulk drug substance. They compared a microcarrier-based perfusion strategy to a fed-batch strategy generating a tonne of mAb with an expected titre < 1 g/L. Bosch et al. (2008) found that the perfusion manufacturing strategy offered savings in COG/g due to the 10-fold increase in productivity relative to the fed-batch bioreactor. The analysis did not consider the consequences of bioreactor failure and focused on a single scale of production. Lim et al. (2005, 2006) compared a spin-filter perfusion to a fed-batch strategy under uncertainty, for an output of 50 kg/year and a titre of 1 g/L. No significant difference in COG/g was seen at this scale in the deterministic analysis. However, the authors used a stochastic analysis to demonstrate the reduction required in the failure rate of the spin-filter perfusion strategy for it to compete with the fed-batch strategy at this scale.

Previous work has either focused on the impact on the cell culture stages rather than the whole process, or on a particular scenario in terms of titre and scale of operation, or not accounted for failure and uncertainties. Furthermore, in recent years new perfusion technologies have emerged,

with the promise of more cost-effective processing and improved environmental performance. Hence this article presents a decision-support tool to evaluate fed-batch and continuous perfusion cell culture technologies, with the ability to rapidly assess the impact on downstream processing, equipment sizing and process economics across a range of scales and titres under uncertainty. An initial deterministic analysis compares fed-batch strategies to both a first generation perfusion system (spin-filter) and second generation perfusion system (ATF) whilst considering the impact of single-use bioreactors. A stochastic analysis was also conducted to understand the impact of failure and its consequences on the robustness of the manufacturing strategies. The paper also takes into account the qualitative concerns associated with the adoption of perfusion technologies and combines these findings with the economic and environmental conclusions to give an overall manufacturing strategy ranking.

Materials and Methods

Model Definition

To successfully evaluate both batch and pseudo-continuous biopharmaceutical unit operations, a simulation tool was required that was capable of optimizing the sizing strategy of each manufacturing scenario whilst accounting for resource constraints and uncertainty in a dynamic environment. The suitability of commercially available bioprocess simulation packages to achieve this was investigated. These packages comprise mainly spreadsheet-based static models that can generate cost estimates but that are unable to capture the impact of resource constraints and the resulting delays in manufacturing (Farid, 2007; Paz and Puich, 2004; Shanklin et al., 2001), a critical feature required for this analysis. Furthermore, they did not enable the specific scheduling features of perfusion culture to be modeled. More specifically, it was not possible to capture the scheduling interactions between the generation of daily perfusion harvests and the subsequent pooling operations prior to batch purification operations as well as the scheduling consequences of failure events occurring at random times during the perfusion cultures. On the other hand, discrete-event simulation languages were found to offer the flexibility to build dynamic models that had all these required capabilities. Hence, a stochastic dynamic decision-support tool was built to capture the specific scheduling challenges with perfusion cultures, as well as the impact of resource constraints, uncertainties and failure events.

The tool comprised a MySQL database (MySQL AB, Uppsala, Sweden) linked to a discrete-event simulation engine, Extend (Extend v6, Imagine That! Inc., San Jose, CA), capable of modeling the operational, economic, environmental, and risk features associated with biopharmaceutical manufacture. The tool was based on a hierarchical data structure outlined by previous UCL

work (Farid et al., 2000, 2005, 2007; Stonier et al., 2009) that breaks down a manufacturing campaign into individual unit operation subtasks at the lowest level. However, the tool required specification of a new level in the form of sub-batches so as to capture the fact that a perfusion reactor generates multiple sub-batches in the form of daily perfusate harvests, which are then subsequently pooled into larger sub-batches prior to further purification.

The tool's simulation engine adopted a router based approach (Stonier et al., 2009) to automatically adapt to different process sequences specified in the database, differing from the conventional linear structure of process models seen in commercial bioprocess simulation packages. A pre-calculation and optimization module was created to run prior to the simulation of a particular scenario so as to assess all the possible equipment sizing strategies before selecting the optimum process configuration (e.g., $1 \times 6,000$ L vs. $3 \times 2,000$ L bioreactors) for the given demand in terms of both production time and cost. Alongside equipment scaling the pre-calculation module also evaluated whether single-use technologies could be implemented in the form of single-use bioreactors and product holding bags, based on the process volumes generated and utilized by each unit operation.

In addition to capturing critical path manufacturing operations, such as cell culture and chromatography, it was important to model and trace the cleaning of process equipment and vessels. This allowed a more comprehensive representation of water and consumable usage to be captured for a specific manufacturing strategy, and hence the potential environmental impact of a process to be addressed alongside the usual economic feasibility and operational impact.

Batch failures were modeled by generating random numbers to represent the percentage chance of failure. The resulting percentage value was then compared to the probability of a given failure event (contamination, filter fouling) for the cell culture operation. If the value was less or equal to the probability of a failure event a batch failure event would be triggered. The consequences of a failure event (see Table IV) were then accounted for on the designated day of occurrence, by either triggering a rescheduling of the fermentation schedule if possible or leaving the equipment idle until the next scheduled batch. Failure events linked to in-process filtration operations led to a yield loss and processing time delay. Uncertainties in input variables such as titre were captured using a function to generate values within a triangular distribution.

The key economic metrics calculated by the tool were the capital investment and the cost of goods per gram (COG/g). The capital investment was determined using the Lang factor method as a function of the total equipment purchase cost (TEPC). The COG/g comprised the annual direct (materials and labor) and indirect (depreciation and facility-dependent overheads) operating costs divided by the annual product output as detailed in Farid (2007). The key environmental metric calculated by the tool was the E-factor (Sheldon,

2007), defined as the total mass of water or consumables used by the manufacturing strategy divided by the total mass of product. The tool's quantitative outputs were combined with qualitative operational metrics such as ease of validation using a multi-attribute decision making (MADM) technique (Hwang and Yoon, 1981) as described in the next section.

Multi-Attribute Decision-Making Methodology

The weighted sum method was used to reconcile economic, environmental, and operational outputs so as to identify the most preferred alternative for scenarios with different weightings assigned to each of these categories. Table II lists all the attributes considered in the MADM analysis. The values of the attributes under economic and environmental feasibility were derived from the simulation tool. The attributes were ranked in order of importance, where a higher ranking indicates an attribute of greater significance. For example, the COG/g was deemed to be slightly more important than initial capital investment required to construct a new facility. The attributes representing the environmental feasibility were ranked equally because the environmental impact of water and consumable usage was deemed to be equally disadvantageous to the environment. The operational feasibility scores and weightings were obtained through a survey questionnaire sent to industrial experts with experience operating both fed-batch and perfusion culture. The participants were asked to rank the cell culture technologies against each of these qualitative attributes: ease of control/operation, ease of validation (time/effort), ease of development (time/effort), operational flexibility and batch-to-batch variability. All attribute values were standardized (Deb, 2008; Triantaphyllou, 2000) to convert them to a common dimensionless scale between 0 and 100. The relative importance of the total weighted economic, environmental, operational scores in the decision making process was captured using a set of combination ratios (dimensionless weight values) whose sum equals one (Deb, 2008). The overall aggregate strategy score S_j is generated by the weighted sum method, using the following

equation:

$$S_j = \left| \frac{\sum_{i=1}^n r_{ijeconomic}}{n} \times R_1 \right| + \left| \frac{\sum_{i=1}^n r_{ijenvironmental}}{n} \times R_2 \right| + \left| \frac{\sum_{i=1}^n r_{ijoperational}}{n} \times R_3 \right| \tag{1}$$

where R_1 , R_2 , and R_3 represent the economic, environmental, and operational combination ratios, respectively.

Case Study

The tool was used to compare commercial mAb facilities using three culture systems: fed-batch (FB) culture, spin-filter perfusion (SPIN), and alternating tangential flow (ATF) perfusion. The case study explored the trade-offs between the higher productivities, and hence smaller upstream and downstream capacities, versus the higher cell culture failure rates caused by filter fouling and contaminations with perfusion systems relative to FB systems. Commercial manufacture of highly successful mAbs can require annual demands of 1,000 kg/year (Kelley, 2007; Rodrigues et al., 2010). An analysis of drug substance demand for 15 mAb and Fc-fusion products by Kelley (2009) demonstrated that annual kilogram output ranged from 1,200 to <100 kg/year, with a median annual kilogram output of 200 kg/year. This case study therefore compared each of the cell culture technologies over a range of scales of production from 100 to 1,000 kg/year so as to explore the rankings over the entire design space. Cell culture titre will also have a large effect on the results of any process comparison (Farid, 2009b). In the past mAb titres reached only mg/L values, but now the norm is 2–3 g/L, with processes in development with reported titres of 5 g/L and higher (Kelley, 2009). Titres of 10 g/L or higher could become the norm in the next 5–10 years. To capture the industry's current and future titre capabilities the case study

Table II. Attribute grouping and ranking.

Attribute field	Attribute name	Rank	FB	SPIN	ATF
Economic feasibility	Cost of goods per gram	2	39	45	31
	Initial capital expenditure	1	88	68	48
Environmental feasibility	Water E-factor rating	1	6,300	9,900	6,500
	Consumable E-factor rating	1	0.2	23.9	19.1
Operational feasibility	Batch-to-batch variability	5	5	9	6
	Ease of control/operation	4	3	9	6
	Operational flexibility	3	3	9	8
	Ease of development	2	3	9	7
	Ease of validation	1	3	9	8

Economic and environmental scores (low = best, high = worst), operational scores (3 = best, 9 = worst)

investigated the following titre values: 2, 5, and 10 g/L. The key metrics used to evaluate the merits and limitations of the three strategies were: COG/g, capital investment, E-factor and operational feasibility.

Key Assumptions

Figure 1 illustrates the process flowsheets for the production of mAbs using the three culture technologies, FB, SPIN, and ATF. Each cell culture operation fed into a single purification train regardless of the number of cell culture operations. When multiple reactors were required to achieve the desired annual demand, two different scheduling approaches were used. The FB process staggered any additional reactors, and altered the scale of the purification train according to the revised times between harvests. In contrast, the SPIN and ATF operations operated the additional reactors in parallel, where the resultant daily perfusate harvests were pooled into a single daily harvest prior to Protein A capture. This combining of different reactor harvests was made possible by inoculating each

reactor from the same seed reactor. Figure 1 also highlights the differences between the primary recovery operations in each flowsheet where the FB process requires the most steps for cell removal and liquor concentration in contrast to the ATF process that does not require any given its external hollow fiber filter. The purification train was based on a generic mAb purification platform using three orthogonal chromatographic steps with intermediate filtration and viral clearance steps (Farid, 2006; Kelley, 2007; Liu et al., 2010). The perfusion culture purification train also includes a pooling operation, where the daily Protein A chromatography eluates were pooled into a larger volume, before being released for further purification. Although, it is possible to locate the pooling operation pre-Protein A and post cation-exchange chromatography, the current commercial manufacturing norm is post-Protein A pooling due to the benefits of operating a smaller highly utilized Protein A column (Wojciechowski et al., 2007).

Table III lists a number of key assumptions that differ between the processes, including the overall process yield, which is lowest for the FB process due to the extra centrifugation and depth filtration operations. The FB

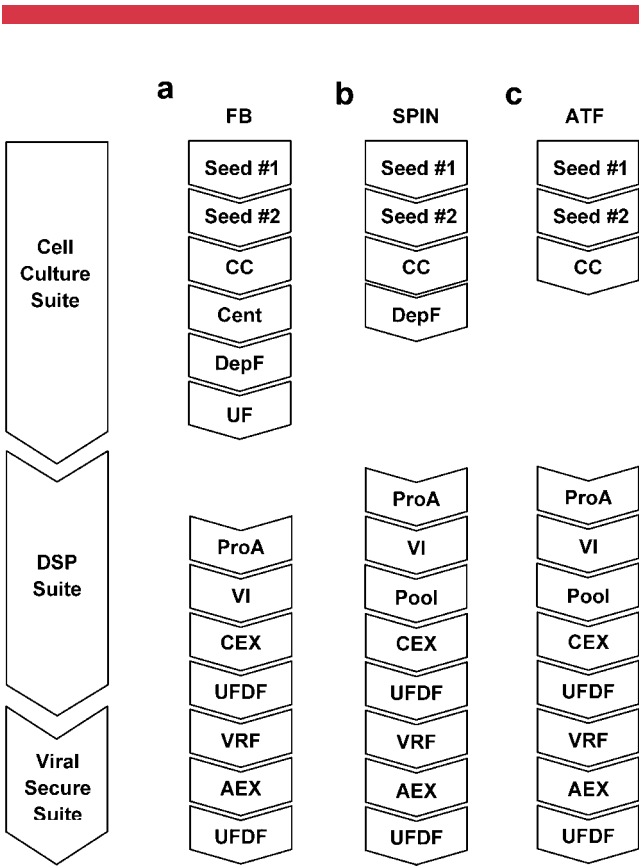


Figure 1. Case study process sequences and suite configuration for (a) the fed-batch (FB), (b) the spin-filter (SPIN), and (c) the alternating tangential flow (ATF) process. CC, cell culture; Cent, centrifugation; DepF, depth filtration; UF ultrafiltration; ProA, Protein A chromatography; VI, virus inactivation; Pool, daily perfusate volume pooling; CEX, cation exchange chromatography; UFDF, ultrafiltration/diafiltration; AEX, anion exchange chromatography; VRF, virus retention filtration.

Table III. Key assumptions for the fed-batch, spin-filter and ATF processes.

Step-specific data			
Variable	FB	SPIN	ATF
Input			
Cell culture time (days)	12	60	60
Max VCD (million cells/mL)	10	15	50
Max bioreactor volume (L)	20,000	2,000	1,500
Max perfusion rate (vv/day)	—	1	1.5
Calculated			
Process yield	65%	68%	69%
Annual # batches	22	5	5
Product concentration (g/L)	2–10	20% FB	45% FB
Volumetric productivity (mg/L/day)	210–1,060	1.6 × FB	5.3 × FB
Annual bioreactor capacity required	1 × FB	1/9 × FB	1/29 × FB
Installed bioreactor capacity	1 × FB	1/2 × FB	1/6 × FB
Grams of product per liter of media	1.2–6.3	19% FB	44% FB
General cost data			
Equipment	Investment Cost (\$)	Consumable Cost (\$)	
Single use bioreactor ^a			
200 L	88,000	4,200	
500 L	98,000	5,460	
1,000 L	110,000	8,260	
2,000 L	175,000	9,800	
Perfusion device ^b			
Spinfilter	35,000	N/A	
ATF 4 system	30,000	714	
ATF 6 system	90,000	3,570	
ATF 8 system	130,000	7,140	
ATF 10 system	180,000	16,300	

^aInvestment cost includes disposable bioreactor support vessel, agitator motor, gas and fluid pumps; consumable cost includes bioreactor bag.

^bInvestment cost includes filter unit, controller and auxiliary media pump; consumable cost includes filter.

Table IV. Monte Carlo assumptions.

Process event	p (failure %)	Consequence
Fed-batch culture contamination	1	Batch loss
Spin-filter culture contamination	6	Batch loss and discard two pooled perfusate volumes
Spin-filter filter failure	4	Batch loss and no pooled volumes are discarded
ATF culture contamination	6	Batch loss and discard two pooled perfusate volumes
ATF filter failure	2	Replace filter and discard next 24 h of perfusate
In process filtration failure—general	5	4 h delay and 2% yield loss
In process filtration failure—post viral inactivation	20	4 h delay and 2% yield loss

culture step ran for 12 days in contrast to the SPIN and ATF cultures which ran for 60 days. A typical maximum cell density for FB cultures of 10 million cells/mL, was assumed. The SPIN culture can generate cell densities of 20 million cells/mL but to maintain the longer culture duration in this case study, a cell density of 15 million cells/mL was selected to prevent premature cell culture termination due to filter fouling. The ATF system is able to reach cell densities of 100+ million cells/mL (Carstens et al., 2009) according to the vendor (Refine Technology), however, to maintain a stable product quality and feasible perfusion rate a more conservative cell density of 50 million cells/mL was used (personal communication with Christoffer Bro and Jacob Jensen of CMC Biologics, Copenhagen, DK). The differences in maximum cell densities result in very different volumetric productivities, with the SPIN and ATF processes offering 1.6- and 5.3-fold increases in productivity, respectively. The cell culture technologies also differ in terms of scalability; FB reactors can reach 20,000 L, whereas SPIN systems peak at 2,000 L and the ATF system at 1,500 L due to the limitations in the maximum perfusion rates currently achieved. The smaller reactor volumes employed by the perfusion systems potentially allow the use of single use bioreactor (SUB) technology. Due to the use of an internal cell separation device in the spin-filter perfusion system, it is not able to use SUBs. In contrast, the ATF system can successfully employ SUBs in all scenarios due to its use of an external cell separation device. The fed-batch system is capable of employing SUBs for reactor volumes below 2,000 L. For the standard fed-batch base case, the use of SUBs was restricted to the seed train where appropriate and stainless steel bioreactors were used for the final production stage as is common industry practice at present for commercial production. In addition, the impact of using SUBs for production bioreactors operated in fed-batch mode was also explored by using new concept facility designs involving multiple smaller SUBs operated in a staggered fashion.

Monte Carlo Analysis Assumptions

Table IV summarizes the probability distributions assigned to the key uncertainties along with the consequences of each failure event used when performing the stochastic cost analysis with Monte Carlo simulations. The probability of cell culture contamination was calculated for each strategy assuming that each addition to the reactor had a 1 in 1,000

chance of causing contamination. The FB strategy had a total of ten reactor additions (initial media fill and nine feeds) and therefore had a 1% probability of batch contamination. In contrast the perfusion strategies had approximately sixty additions due to the daily media exchanges and therefore had ~6% probability of contamination. The risk of equipment failure due to filter fouling in the perfusion strategies was also considered. The SPIN strategy had a higher probability of filter failure compared to the ATF strategy due to the use of an internal filter that actually relies on a degree of surface cell growth to prevent excessive cell carryover into the perfusate. The probability of filter fouling increases with duration of the perfusion cell culture (Deo et al., 1996; Vallez-Chetreau et al., 2007). The probability of filter failure is shown as a percentage in Table IV but when the tool selects a filter failure scenario, the occurrence of the failure is weighted to occur at the latter stages of the cell culture duration. Batch-to-batch titre fluctuations in cell culture were also captured as they can have a significant impact on the mass of antibody generated impacting the purification operations and hence annual kilogram output and COG/g. Typical titre fluctuations at commercial scale are $\pm 20\%$. This was implemented by applying a triangular distribution to the cell line productivity for each batch with the minimum and maximum values being $\pm 20\%$ of the most likely base value. A further common uncertainty is failure of in-process filtration (IPF) operations associated with every product tank/bag fill or draining protocol. The majority of IPF failures occur due to the blocking of the sterile filters, hence the probability of IPF failure is significantly higher after viral inactivation due to the increased probability of aggregate formation due to the low pH holding conditions.

Table IV also highlights the consequences of a failure event, which vary depending on the cell culture strategy. For the FB scenario a contamination resulted in the loss of the whole batch. However, for the perfusion strategies material harvested prior to the contamination event could be processed apart from the two latest pooled perfusate volumes that were discarded. The perfusion strategies are also prone to filter failure which would also halt the cell culture upon a failure event. The use of an external filter by the ATF strategy meant the filter could be replaced and the cell culture resumed, after discarding the subsequent day's perfusate volume due to high levels of HCP. The SPIN strategy employs an internal filter which cannot be replaced

mid-culture resulting in the halting of the cell culture, but allows all prior harvested material to be processed. When a failure event occurred, the discrete-event simulator triggered the start of the subsequent planned batch so as to prevent idle time and poor facility utilization, with a time lag of approximately 2 weeks for the production reactor inoculum to be generated. This had little effect on the fed-batch scenario since the batch interval was also approximately 2 weeks, meaning the next planned batch would be available prior to any replacement batch. In contrast for the perfusion systems with long culture durations, the rescheduling could potentially allow a replacement batch to be processed depending on the timing of the failure event. In the event of an IPF failure event, the process was delayed by 4 h and 2% of the product material was lost. Advice from industrial experts was solicited to sanity check the validity of these risks and the resulting consequences on production.

Results and Discussion

The tool was used to assess how the cost-effectiveness of the fed-batch and perfusion strategies changes across a range of titres and scales of production. This was initially carried out

deterministically and then expanded to account for the impact of equipment failures and uncertainties in performance. The economic outputs were then considered alongside operational and environmental metrics using a multi-attribute decision-making technique.

Deterministic Cost Comparison

COG/g Comparison Across Scales

Figure 2 shows the COG/g for the FB, SPIN, and ATF strategies for the 5 g/L scenario across a range of scales of production (100, 500, 1,000 kg/year). The analysis suggests that although the SPIN strategy offers similar COG/g values to the FB strategy at the smallest production scale of 100 kg/year, it becomes less economically attractive at the higher production scales. In contrast, the ATF strategy is seen to offer cost advantages across all production scales of ~20% in the 5 g/L scenario.

A closer examination of the COG/g breakdowns highlighted in Figure 2 reveals how changes in the relative importance of different cost categories influence the cost-benefit rankings. The trade-off between the lower indirect

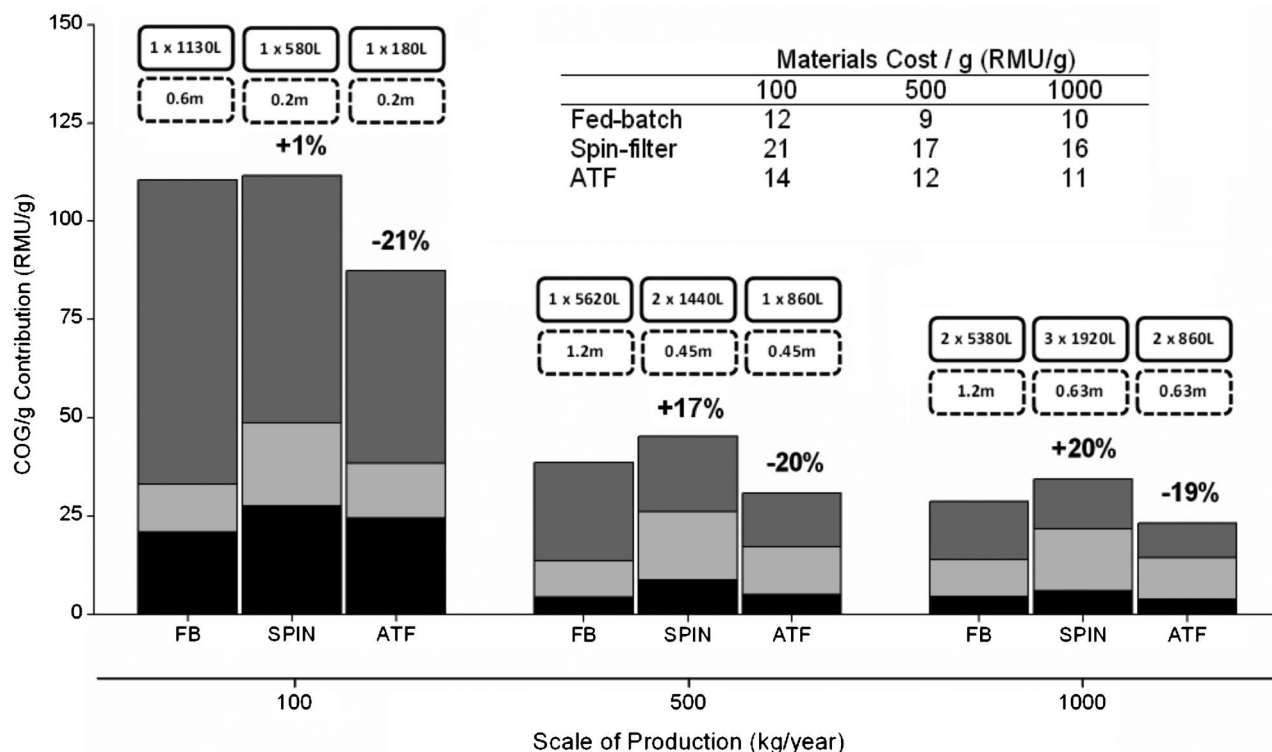


Figure 2. A comparison of the cost of goods per gram on a category basis for labor costs (■), direct material costs (■), and indirect costs (■) between the fed-batch process (FB), the spin-filter process (SPIN) and the alternating tangential flow process (ATF) over a range of scales of production for an equivalent fed-batch titre of 5 g/L, where the percentage difference is relative to the fed-batch process. The optimal sizing strategy for each process is indicated in the boxes above each bar highlighting the number and scale of bioreactor(s) (–) and the column diameter for the Protein A chromatography step (–) across a range of scales of production.

costs and higher material costs with the SPIN strategy relative to the FB strategy is dependent on the production scale. At the smallest production scale, changes in material costs can be seen to have much less influence due to the dominance of the investment-driven indirect costs, as has been echoed in other reports (Farid, 2007, 2009a). Across all production scales, the material costs per gram decrease at a slower rate than the other leading cost contributors as illustrated in the embedded table in Figure 2 and the SPIN strategy exhibits a ~ 1.8 -fold increase in material costs (16–21 RMU/g) relative to the FB strategy. This difference in material costs becomes increasingly more significant at the higher production scales as material costs represent a much higher proportion of the COG/g, contributing to the reduced competitiveness of the SPIN strategy. In contrast, the ATF strategy is able to compete across all scales since its superior cell density and hence volumetric productivity coupled with a smaller highly utilized purification train results in significantly larger savings in indirect costs ($\sim 40\%$) combined with only a ~ 1.2 -fold increase in material costs. This translates into overall savings irrespective of the dominance of either indirect or material costs at either extremes of the production scales.

The tool was also used to identify the material cost drivers for each strategy. Figure 3 presents a detailed percentage breakdown of the COG/g, for all three strategies in the scenario at 5 g/L and a 500 kg/year production scale, on a category basis with particular emphasis on the key material costs. The higher material cost contributions in the perfusion systems can be attributed primarily to the higher usage of culture media, product-holding bags, and SUB liners (for ATF systems). It is interesting to note that although the dominant material cost shifts from chromatography resins (9% COG/g) in the FB strategy to culture media (12% COG/g) in the SPIN strategy, this shift in dominance is not seen in the ATF strategy. This is in spite of the fact that increases do occur in other cost categories. This can be attributed to the more productive ATF systems utilizing comparably less media than the SPIN strategy; hence the increase in media cost is not substantial enough to outweigh the high costs associated with the chromatographic resins.

The impact of new concept facility designs based on multiple staggered single-use bioreactors operated in FB mode was also investigated. A new concept SUB facility (FB-SUB) with a 500 kg/year scale of production and a 5 g/L titre would require four 1,500 L staggered SUBs with a harvest frequency of 3 days, compared to a single 5,600 L stainless steel bioreactor with a harvest frequency of 12 days. Capital investment estimates ($\text{TEPC} \times \text{Lang factor}$) required the derivation of a suitable Lang Factor (24) for the SUB-based cell culture suites so as not underestimate their infrastructure costs (see Appendix Table AI). Despite a shift in the COG breakdowns to a more evenly spread distribution across indirect, material and labor costs (Fig. 3d), no significant difference was seen in the magnitude of COG values between the FB and FB-SUB strategies.

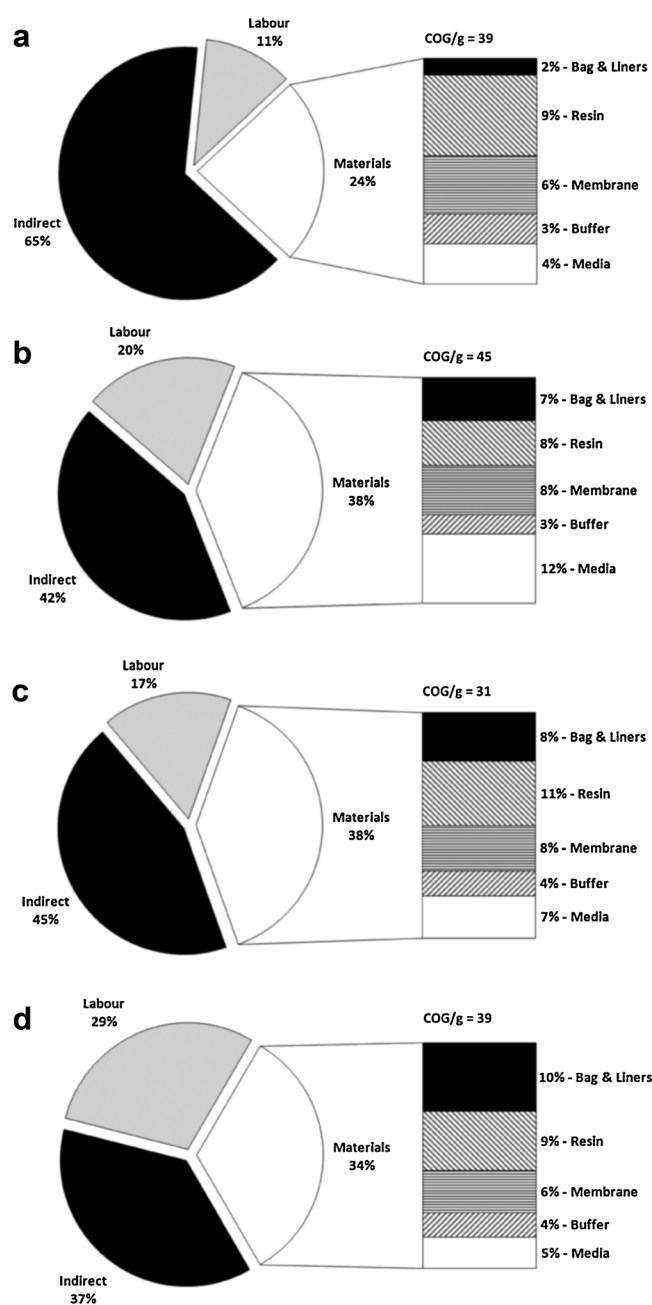


Figure 3. A comparison of cost of goods per gram with a detailed breakdown of material costs on a category basis for (a) the fed-batch process, (b) the spin-filter process, (c) the alternating tangential flow process, and (d) the concept fed-batch SUB process, for a 500 kg/year scale of production and an equivalent fed-batch titre of 5 g/L. The COG/g values for each process are also indicated.

Consequently, FB strategies with multiple SUBs are still unable to compete with the ATF COG savings.

Key Economic Metrics Across Scales and Titres

The impact of both scale of production (100, 500, 1,000 kg/year) and titre (2, 5, 10 g/L) on the competitiveness of the

three strategies was investigated. The contour plot in Figure 4a shows the results for the SPIN strategy relative to the FB strategy when expressed as percentage change. The figure highlights that the SPIN strategy is only able to compete with the FB strategy at either the low production scales (100 kg/year), irrespective of titre, or at the high titres of 10 g/L across all the scales (with a $\pm 10\%$ difference). Both these sets of conditions place low demands on bioreactor capacity where the savings on investment-related indirect costs dominate. Hence, the benefits of installing a single bioreactor train that is typically half the size of the FB strategy is balanced against the higher material costs. In contrast, as the scale of production (kg/year) increases, coupled with decreases in titres, the SPIN strategy becomes increasingly unattractive. This effect is particularly pronounced at the lowest titre (2 g/L) and the highest scale of production (1,000 kg/year), where the SPIN strategy's COG/g reaches double that of the FB strategy. This can be attributed to the need for a high number (8) of SPIN reactors to cope with the large-scale of production, given their scale limitations combined with the larger bioreactor capacities required at lower titres. This negates any potential advantages offered by having lower total installed bioreactor capacities with the SPIN strategy relative to the FB strategy. Instead the results generated by the tool indicate that the difference in capital investment and the related indirect costs ($\times 1.4$) as well as labor costs ($\times 3$) between the SPIN and FB strategies become exaggerated. These increases coupled with the higher material costs in perfusion strategies illustrate how the SPIN strategy loses its competitive advantage under high bioreactor capacity demands that result in scale-out to multiple bioreactors.

In contrast, the impact of increasing titre across the different scales of production does not significantly affect the cost-effectiveness of the ATF strategy (Fig. 4b) in the same manner as the SPIN strategy. The superior volumetric productivity of the ATF strategy relative to the FB strategy

(5.3-fold higher), leads to significantly smaller bioreactor volumes compared to the SPIN strategy and prevents the rapid increase in bioreactor number and COG/g values seen. This translates into an installed reactor capacity that is approximately six times smaller than the FB strategy in comparison to the SPIN strategy's ability to only offer half the installed reactor capacity of the FB strategy. The installed bioreactor capacities are derived from the annual bioreactor capacities divided by the number of batches. Given that the perfusion strategies only operated five batches annually compared to twenty-two for the FB strategy, the ATF strategy offers a 29-fold decrease in annual reactor capacity compared to only a ninefold decrease with the SPIN strategy. The superior productivity of the ATF strategy coupled with a small highly utilized purification train, allows the ATF strategy to offer capital investment savings of $\sim 40\%$ and COG/g savings of $\sim 20\%$ across all titres and scales of production compared to the FB strategy.

These results assumed a fivefold difference in maximum viable cell density between the ATF and FB systems. The tool was also used to identify the critical threshold value for the maximum viable cell density in the ATF system for it to remain competitive across different scales of production and titres (Fig. A1). The tool predicted that even if the ATF cell density dropped such that only a threefold increase in cell densities was achieved relative to the FB system (Fig. A1b), the ATF system would still offer COG/g savings compared to the FB strategy across the different combinations of production scales and titre, except for the worst case combination of a low titre (2 g/L) and high production scale (1,000 kg/year). Below this threshold cell density, the ATF system's ability to compete with the FB system is reduced to a subset of the combinations of titres and scales of production investigated. For example, if the ATF system was only able to achieve a twofold higher cell density (Fig. A1a), it would be economically attractive at either the low production scale (100 kg/year) across all the titres or at

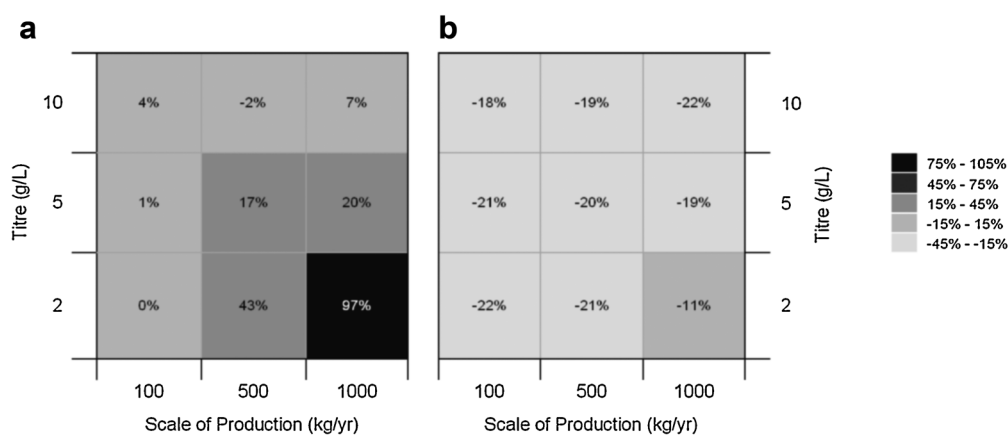


Figure 4. Contour plot showing the impact of scale of production and titre on the percentage difference in COG/g relative to the fed-batch process for (a) the spin-filter perfusion process and (b) the alternating tangential flow (ATF) perfusion process. The processes are resized for each combination of scale of production and titre.

the high titre (10 g/L) across all the scales. This is similar to the trend observed with the SPIN system with its 1.5-fold higher cell density relative to FB; however, the ATF system's drop in competitiveness is less severe than that seen with the SPIN system due to the use of single-use bioreactors and the removal of a depth filtration step.

Stochastic Cost Comparison

The initial deterministic cost comparison highlighted the overall economic advantages offered by operating the ATF strategy. The study was extended to include the perceived risks associated with perfusion based strategies, namely cell culture failure attributed to either contamination or equipment failure due to filter fouling over the long culture durations. The Monte Carlo simulation technique was used to characterize the variability in the kilogram output and COG/g values caused by fluctuations in failure rates and titre. The following discussion highlights the key findings from this analysis and assesses the robustness of the FB and perfusion strategies.

Expected Scenario Outputs

Figure 5 shows the expected kilogram output and COG/g values for the FB, SPIN, and ATF strategies under uncertainty at a 500 kg/year scale of production and a titre of 5 g/L. Figure 5a depicts the expected annual output achieved by each cell culture strategy, showing every kilogram output value recorded (thin lines), the frequency that the values occur (line plot). The FB strategy is the most robust strategy as indicated by the narrower spread of the frequency values (410–490 kg/year). The frequency of annual output values for the FB scenario has a distinct shape which was found to relate directly to the number of batch failures in any given year of production. This is highlighted in Figure 5c where the biggest bar represents no batch failures and the subsequently smaller bars represent the occurrence of one and more batch failures per year. The robustness of the FB strategy can be attributed to the low probability of cell culture failure (1%) and the high number of batches (22 batches/year) lessening the effect of titre variation on annual kilogram output. In contrast the perfusion strategies are not as robust as the FB strategy due to their higher probability of failure and fewer number of batches per annum (5 batches/year) amplifying the effect of titre variation on the annual kilogram output achieved. The shape of the frequency plots depicting annual kilogram output is not as defined as the FB strategy either, due to the variability in the timing of a failure event dictating how much material has been processed from the failed batch. The perfusion strategies both have occurrences of kilogram outputs over 500 kg/year due to scenarios when all the titres achieved that year are above average and there were no cell culture failures. The ATF strategy achieves high kilogram outputs more readily due to only 6 in 100 cell cultures runs

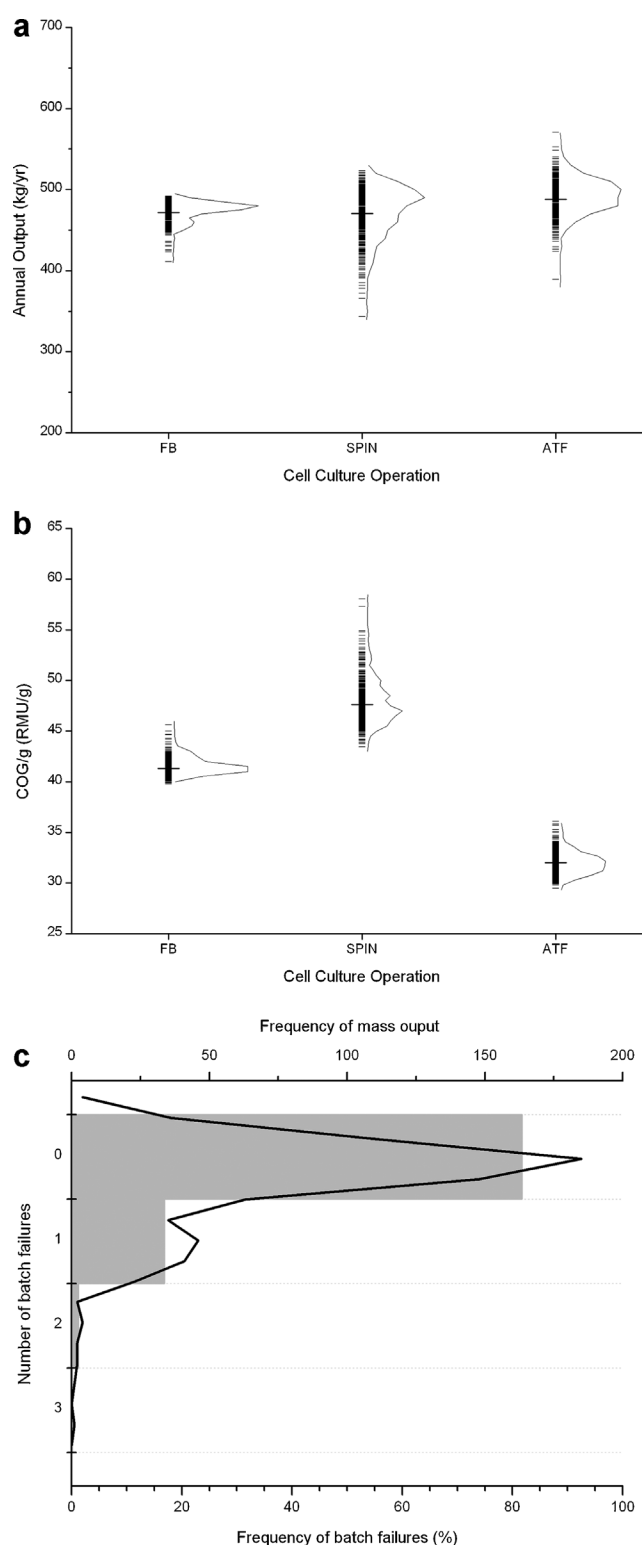


Figure 5. Frequency distribution plots depicting the expected process outputs under manufacturing uncertainty for (a) the expected annual kilogram output, (b) the expected cost of goods per gram, and (c) the number of fed-batch culture failures, for a 500 kg/year scale of production and equivalent fed-batch titre of 5 g/L.

failing compared to 1 in 10 failures for the SPIN strategy. This results in the ATF strategy having a higher expected annual kilogram output of 488 kg/year compared to the SPIN strategy value of 470 kg/year and the FB strategy value of 472 kg/year, even though the SPIN strategy often fails to achieve kilogram output values above 400 kg/year. The *t*-static value for the ATF strategy versus the FB strategy indicates that there is a statistically significance difference.

Figure 5b shows the expected values for the COG/g, where the COG/g frequency plots are a reflection of the annual kilogram output frequency plots, given that a higher output results in better facility utilization and hence a reduced COG/g value. The only exception to this rule occur when the ATF strategy achieves an annual output over 500 kg/year, which is a higher output than the facility design, resulting in a higher COG/g than expected due to the extra costs required to process the extra material, resulting in a frequency plot more aligned with the FB strategy. The ATF strategy further demonstrates its economic superiority by maintaining a ~20% reduction in COG/g with an expected value of 32 RMU/g compared to the FB strategy's expected COG/g value of 41 RMU/g. The ATF strategy is also able to maintain this COG/g advantage over the FB strategy even in its worst-case scenario involving multiple failure events. Overall the expected stochastic COG/g values are ~5% higher than the deterministic values and maintain the economic ranking shown in the deterministic cost comparison, however the stochastic analysis gives a clear indication of the robustness of the manufacturing strategy employed unavailable through deterministic analysis.

Multi-Attribute Decision-Making

This section extends the analysis beyond economic metrics to include the environmental and operational benefits of each strategy.

Environmental Impact Analysis

The tool was also used to capture the water and consumable usage of the strategies to assess the environmental impact of the FB and perfusion strategies across a range of titres and scales of production. E-Factor values were derived for the usage of process water (cell culture media and process buffers), non-process water (CIP buffers and rinse water) and consumables (bags, membranes, and resins). Typical FB manufacturing strategies consume water (process and non-process) from 3,000 to over 7,000 kg water per kilogram product. The cell culture steps consume between 20% and 25% of the total, with the chromatographic operations often surpassing 50% of the total (Ho et al., 2011).

As expected the perfusion strategies have a much higher process water E-Factor value in comparison to the FB strategy, where the higher process water E-Factors can be directly related to the volumetric productivity of the perfusion strategies and their resulting media usage. The

perfusion strategies extra burden on CIP caused by the highly utilized primary clarification operations and purification train for the high number of pooled perfusate volumes, results in the SPIN strategy consuming double the non-process water relative to the FB strategy. In contrast the ATF strategy has a ~30% lower non-process water E-Factor value relative to the FB strategy, due to the ability of the strategy to utilize SUB technology and the removal of a dedicated primary clarification operation. The consumable E-Factor values are highly dependent on the amount of single use technologies employed by the strategies. This effect is particularly pronounced at the smallest scale of production (100 kg/year), where all the manufacturing strategies can successfully employ single-use technologies, resulting in higher consumable E-Factor scores and lower non-process water E-Factor scores. The smaller purification train in the perfusion strategies is ideally suited to such technologies across all the scales of production, leading to the high consumable E-factor scores seen in Table V.

Qualitative Operational Benefits

Qualitative attributes related to the operational feasibility of the strategies were derived from survey responses from industrial experts with both FB and perfusion experience. In terms of attribute weights, the responses highlighted that the batch-to-batch variability and ease of control/operation were considered more important than the other attributes. In terms of rating each strategy against these attributes, Table II indicates that the FB and ATF strategies scored evenly with regard to batch-to-batch variability due to the ability to successfully control the cell culture conditions for the strategies. The FB strategy was the clear favorite with regard to ease of control/operation due to the added complexity associated with the perfusion strategies' cell retention devices and daily feeding operations.

Overall Aggregate Strategy Scores

The results of reconciling the trade-offs between economic, environmental and operational outputs using a single multiple attribute score are reviewed in this section. The key output was the overall aggregate strategy score over a range of combination ratios to reflect the impact of the relative importance of the economic, environmental and

Table V. E-factor scores for water and consumable consumption.

	E-Factor scores (kg/kg product)		
	FB	SPIN	ATF
Process water	4,300–5,050	4,950–9,800	4,700–5,300
Non-process water	1,150–4,950	2,600–12,500 ^a	1,400–6,300
Consumables	0.2–16.1	7–48.8	4.6–37.4

^aHigh spin-filter perfusion strategy water E-Factor scores is a consequence of a high number of reactors employed.

operational scores on the ranking of the manufacturing strategies. Figure 6 depicts the sensitivity of the overall aggregate strategy scores to the economic attribute combination ratios for the FB, spin-filter and ATF strategies at a 500 kg/year scale of production and a titre of 5 g/L. For the scenario shown in Figure 6a, the operational attribute combination ratio was fixed at 0.1 and the environmental attribute combination ratio varied with the economic attribute combination ratio such that the sum of all the combination ratios always remained equal to one. Figure 6a illustrates that when the environmental and economic scores are equally weighted ($R_1 = R_2 = 0.45$), the ATF strategy has the highest aggregate score and therefore would be the preferred manufacturing strategy. If the environmental benefits are considered to be approximately 1.5 times as important as the financial benefits ($R_1 = 0.35$, $R_2 = 0.55$) the FB and ATF strategies are equally attractive and outperform the SPIN strategy. The FB strategy only comes out as the

superior strategy over the ATF and SPIN strategies when the environmental benefits are considered over twice as important as the financial savings. Interestingly, Figure 6b shows that when the environmental attribute combination ratio is fixed at 0.1 and the operational and economic scores are equally weighted ($R_1 = R_3 = 0.45$), the FB and ATF are equally ranked. Figure 6b illustrates that when the economic benefits are considered more important the ATF strategy is the preferred manufacturing strategy, however if the operational benefits are more important the fed-batch strategy becomes the favored manufacturing strategy. These relationships were maintained across all the annual kilogram outputs investigated.

Conclusions

This article presents the development of a dynamic decision-support tool to assist in cost-effective bioprocess design in the presence of uncertainty and multiple conflicting outputs relating to economic, environmental, and operational feasibility. The tool was configured to cope with the continuous nature of perfusion cultures, the consequences of failures as well as the use of single-use bioreactors and bags when the scale was appropriate. The tool was used to provide an in-depth analysis of the potential of mAb facilities based on fed-batch processes compared to 1st (spin-filter) and 2nd (ATF) generation perfusion systems across a range of titres and scales of operation so as to represent different possible scenarios of relevance to industry. The analysis highlighted the underlying cost drivers for each process and identified the robustness of each process along with the root causes for the differences. The derivation of environmental indices not only provided useful benchmarks of E-factors for fed-batch and perfusion processes but also enabled the economic, environmental, and operational outputs to be assessed simultaneously. This was achieved using a multi-attribute decision-making technique that provided a more holistic approach to managing conflicting outputs. The tool's predictions that the spin-filter perfusion strategy struggles to compete on economic, environmental, operational, and robustness fronts at most titres and scales provides insight into its limited use in industrial processes. In contrast, the ATF perfusion strategy is predicted to offer economic benefits that outweigh its lower robustness, even when it achieves cell densities that are only threefold higher than fed-batch strategies for typical titre and demand levels. However, the analysis highlighted that if environmental or operational feasibility (e.g., ease of operation and validation) are considered more important than process economics savings then the fed-batch strategy is found to be preferred. The tool therefore acts as a valuable test bed for assessing the potential of novel strategies to cope with future titres and scales of operation.

The authors would like to thank the following industrialists for sharing their experience with running fed-batch and perfusion sys-

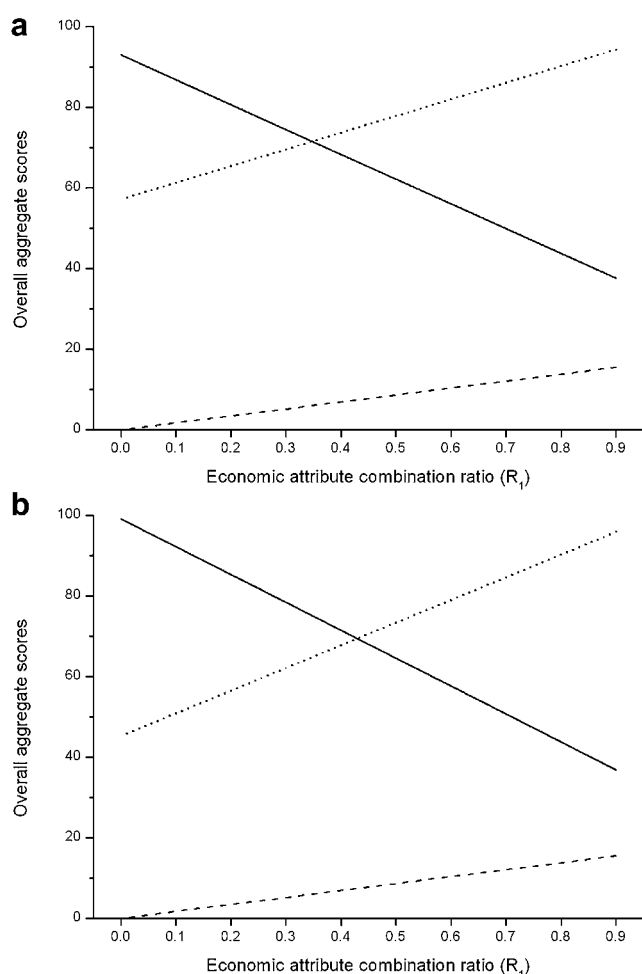


Figure 6. Sensitivity plots portraying the effect of the economic attribute combination ratio (R_1) on the overall aggregate scores when (a) the operational attribute combination ratio is constant and (b) the environmental attribute combination ratio is constant. For the fed-batch (—), spin-filter (---), and ATF (···) processes, for 500 kg/year scale of production and equivalent titre of 5 g/L.

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APPENDIX

Figure A1, Table A1

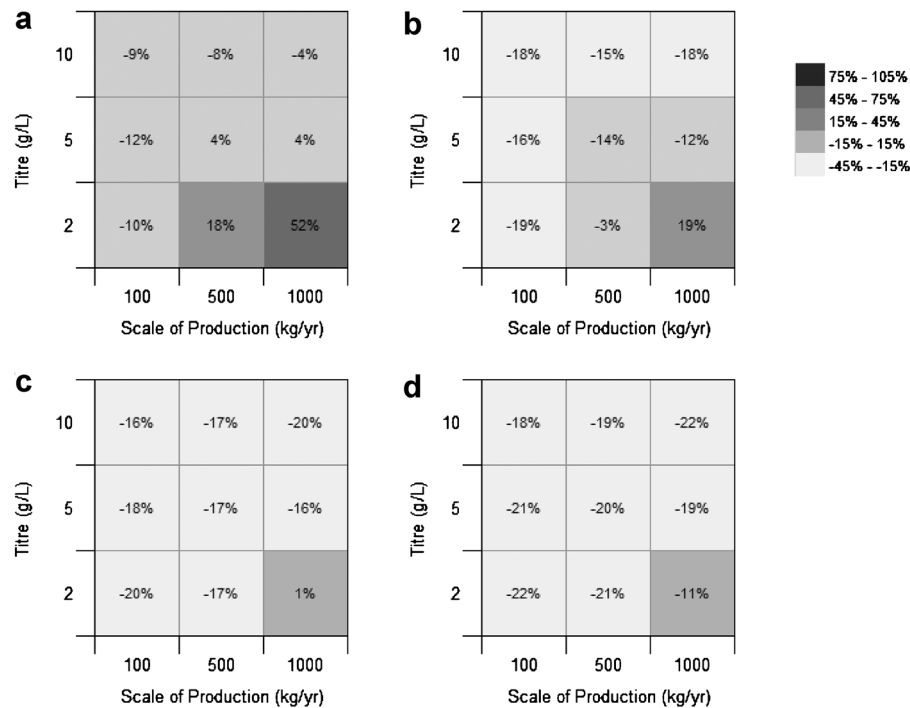


Figure A1. Contour plot showing the impact of the ATF system's viable cell density at different scales of production and titres on the percentage difference in COG/g relative to the fed-batch process for (a) 20 million cells/mL, (b) 30 million cells/mL, (c) 40 million cells/mL, and (d) 50 million cells/mL. The fed-batch system was assumed to achieve a maximum viable cell density of 10 million cells/mL. The processes are resized for each combination of scale of production and titre.

Table A1. Bioprocessing facilities capital investment factors and corresponding "Lang" factors.

Factor description		Conventional/conventional TEPC*	Disposable/conventional TEPC*	Disposable/disposable TEPC
1	Total equipment purchase cost (TEPC; incl. utilities)	1	0.2	1
2	Pipework and installation	0.9	0.3	1.49
3	Process control	0.37	0.37	1.85
4	Instrumentation	0.6	0.4	1.98
5	Electrical power	0.24	0.24	1.2
6	Building works	1.66	1.33	6.64
7	Detail engineering	0.77	0.39	1.93
8	Construction and site management	0.4	0.30	1.5
9	Commissioning	0.07	0.07	0.35
10	Validation	1.06	0.53	2.65
	Contingency factor	1.15	1.15	1.15
	"Lang" factor	8.13	4.73	23.67

*Adapted from Novais et al. (2001).